

Mack Thompson

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EDUCATION

Auburn University

May 2027

B.S. in Computer Science, Minors in Mathematics & Business

GPA: 3.76/4.00

- Competitive Programming Team, Association for Computing Machinery (ACM), TFAS Charity Boxer, ΛXA fraternity

EXPERIENCE

AI Engineer Intern

Incoming Fall 2026

Southern Company

Atlanta, GA

- Incoming AI Engineering Intern working with the CITO to apply NLP and agentic AI systems to cybersecurity workflows.

Cloud Engineer Intern

Incoming Summer 2026

Hewlett-Packard Enterprise

Sunnyvale, CA

- Selected for HPE's Cloud Engineering internship to develop and optimize distributed systems and networking infrastructure in enterprise cloud environments.

Research Fellowship (2x)

Feb 2025 – Present

Auburn University – Computer Vision & Machine Learning

Auburn, AL

- Developing sequence-based ML model on EV battery telemetry to forecast temperature evolution and identify stress.
- Scraped dermatology dataset of 5k+ labeled images, applying augmentation and ETL preprocessing to stabilize training.
- Uncovered misclassification biases linked to skintone after evaluating EfficientNet-B0 with grouped confusion matrices.

Software Engineer Intern

Sep 2025 – Present

OCV Apps – Public safety mobile apps

Opelika, AL

- Eliminated redundant 5K-line JSON manifests across 40+ apps by introducing a master schema with per-app diffs.
- Built a diff/merge engine to reconstruct app-specific configs at runtime, reducing update overhead by ~80%.

Software Engineer Intern

May 2025 – Aug 2025

Kahua Inc. – SAAS for Construction

Alpharetta, GA

- Decreased automated test development time by 30% by enabling code modification without resetting the test driver.
- Improved deployment reliability by expanding an Azure DevOps testing pipeline, automating 20+ key client workflows.

Founder

Jun 2021 – Dec 2024

Summer Sports Camp

Atlanta, GA

- Scaled to 100+ campers & 6 staff each summer, generating \$100K+ in profit before transferring ownership.
- Deployed customer response agent with AWS Pub/Sub, automating 75% of client inquiries & eliminating wait periods.

PROJECTS

AI lifeguard system

Mar 2026

OpenCV, Pytorch

- Partnered with university facilities to build a system for pool safety, integrating hardware, transport, & inference.
- Machined a CAD camera ring & built a multiplexing video pipeline for wide-angle underwater capture and transmission.
- Shipped a lifeguard-facing GUI to stitch camera feeds, run real-time CV detection, and trigger alerts for drownings.

JupJewels.com

Jul 2025 – Sep 2025

React, Shopify, JavaScript

- Developed a custom jewelry builder that became the top-selling product, generating \$4K revenue in 30 days.
- Redesigned storefront using customer feedback, increasing site traffic by 25% and improving conversion rates.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C/C++, C#, HTML/CSS

Frameworks & Libraries: React, Flutter, PyTorch, Pandas, Selenium, NumPy

Tools: AWS, Git, Azure, Node, PowerBI, SQL, Bash, REST, Firebase, Docker, PostgreSQL, SQLite, Excel

Awards: ICPC Southeast regional 3rd Place 2023 & 2025, GHSA AAA Regional 153 lbs Wrestling Champion 2023